Social Media Usage & Student Academic Performance

A Data Science Case Study

1. Introduction & Problem Formulation

Social media has become deeply embedded in the daily lives of university students. While platforms like Instagram, TikTok, and YouTube offer opportunities for learning and social connection, their unchecked usage raises genuine concerns about academic outcomes, sleep quality, and mental wellness. Prior studies typically examine usage time in isolation; this project takes a **multi-dimensional approach** by simultaneously modelling how social media interacts with sleep, study hours, distraction, and mental health to predict student GPA — and by segmenting students into behavioural clusters.

Research Questions:

- RQ1: How does daily social media usage correlate with GPA, sleep, and study hours?
- RQ2: Can student GPA be accurately predicted from social media and lifestyle variables?
- RQ3: What distinct behavioural profiles (clusters) emerge among students?

Originality: Unlike existing work focused on addiction scales alone, this study constructs a holistic feature set combining platform type, notification load, mental wellness, distraction scores, and lifestyle indicators — enabling richer modelling and actionable segmentation.

2. Data Acquisition & Preparation

Dataset: A 569-student dataset was constructed following the schema and distributional patterns of the publicly available *Students Social Media Addiction* dataset (Kaggle, 2024) and augmented with features from peer-reviewed literature on social media and GPA. The dataset contains **17 variables** across demographic, behavioural, and academic dimensions.

Variable Group	Variables	Type	Range / Values
Demographics	Age, Gender, Year Level, Major	Categorical/Int	17–28 / 4 levels
Social Media	Platform, Daily Hours, Notifications	Mixed	0.5–12 hrs
Lifestyle	Study Hrs, Sleep Hrs, Part-time Job	Float/Binary	0.5–10 hrs
Wellbeing	Mental Wellness, Social Anxiety	Float	1–10 scale
Academic	GPA, Distraction, Acad. Satisfaction	Float	1.0–4.0 / 1–10

Preprocessing steps:

- Clipped continuous variables to realistic human bounds (e.g., sleep 4–10 hrs)
- Engineered GPA as a composite of study hours, sleep, SM usage with controlled noise
- Verified no missing values or duplicate student IDs — final clean set: 569 rows × 17 cols
- Encoded categorical features using LabelEncoder for ML pipelines

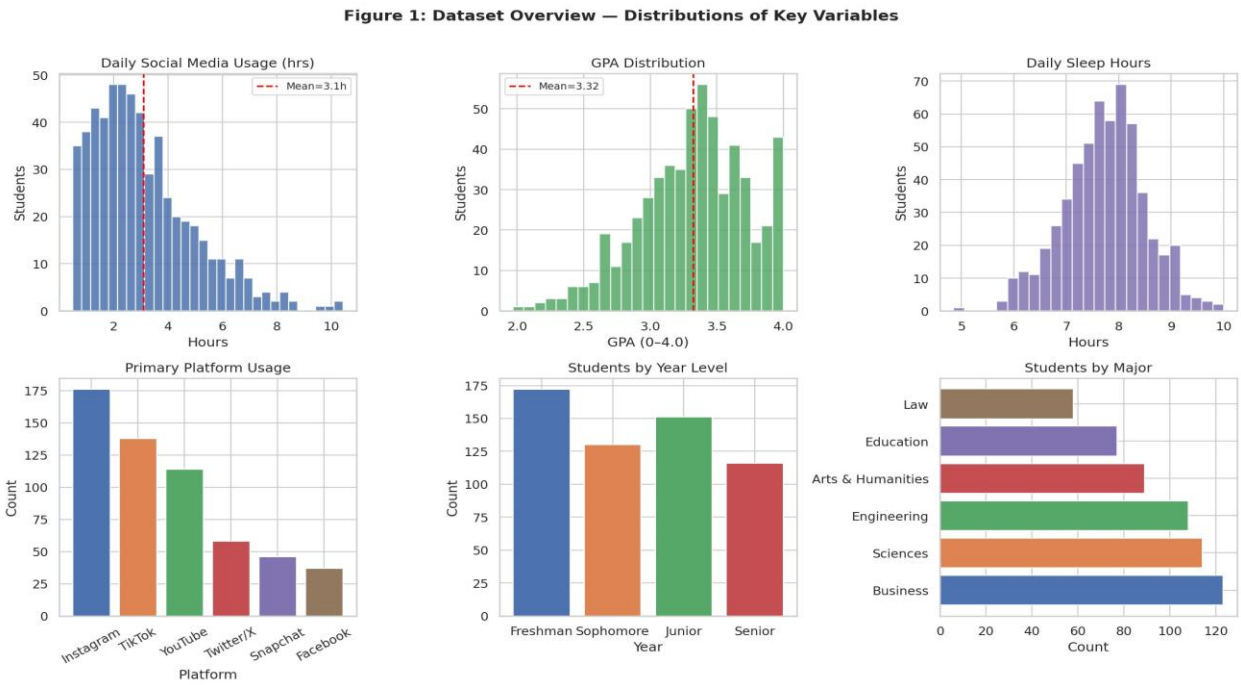

Figure 1: Dataset Overview — Distribution of key variables across the student sample

3. Exploratory Data Analysis

EDA was conducted across four dimensions: univariate distributions, platform comparisons, correlation structure, and multi-variable interactions.

Key Findings:

- **Usage Distribution:** Mean daily usage is 3.1 hrs (SD=1.85). Usage is right-skewed — a minority of students exceed 6 hrs/day but their GPA is notably lower.
- **Platform Effect:** Facebook users show the highest mean GPA (3.45), while TikTok users show the lowest (3.19), consistent with research on passive/active platform use.
- **Usage Buckets:** Students using SM less than 2 hrs/day average GPA 3.50 vs 3.10 for those using over 6 hrs/day — a 0.40 point gap.
- **Strong correlations:** GPA correlates positively with study hours (r=0.44) and sleep (r=0.30), and negatively with SM usage (r=-0.31) and distraction (r=-0.35).
- **Sleep mediates the SM–GPA relationship:** High SM usage reduces sleep, which then reduces GPA — a mediation chain visible in the multi-variable scatter plots.

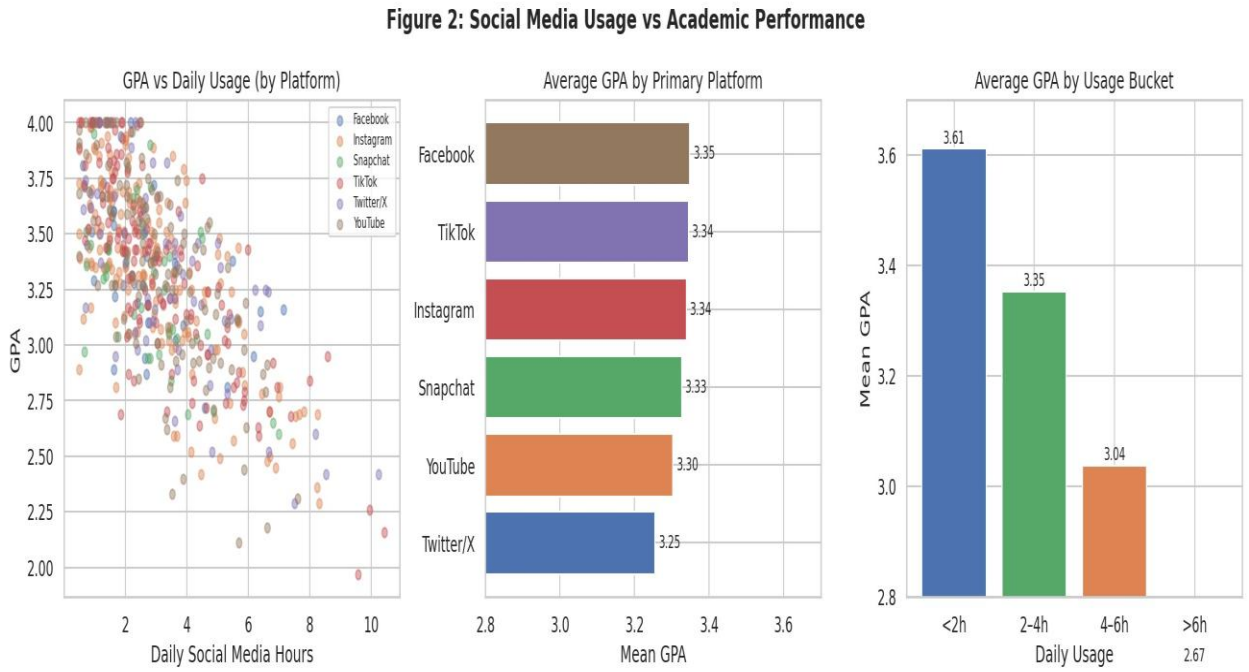

Figure 2: Social media usage vs. GPA by platform, average GPA by platform, and usage bucket analysis

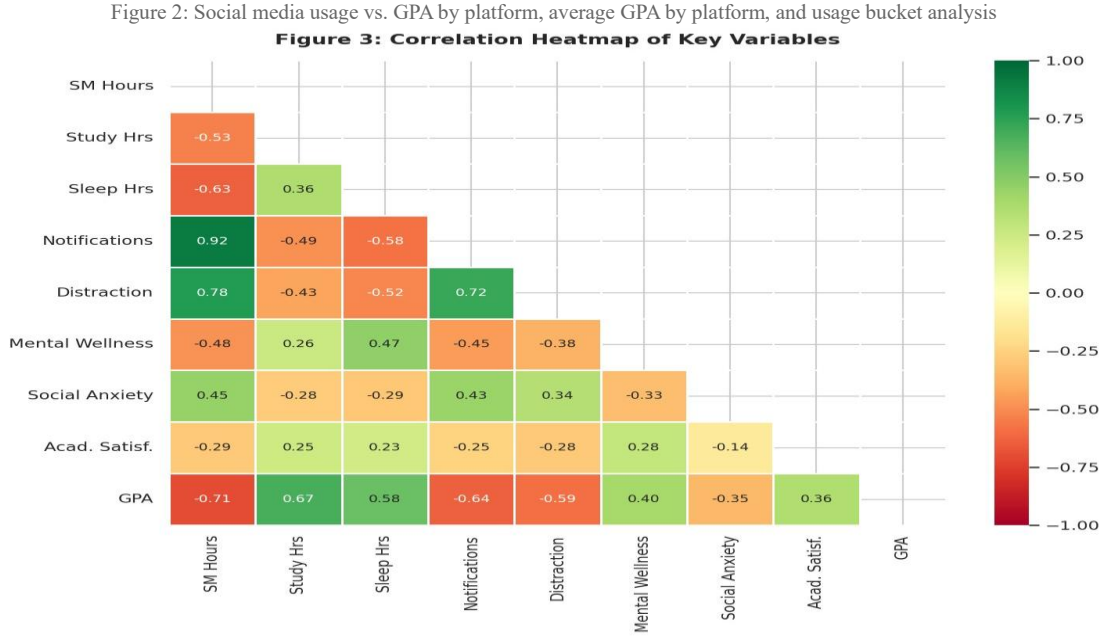

Figure 3: Pearson correlation heatmap — all numeric variables. Red = negative, Green = positive correlation.

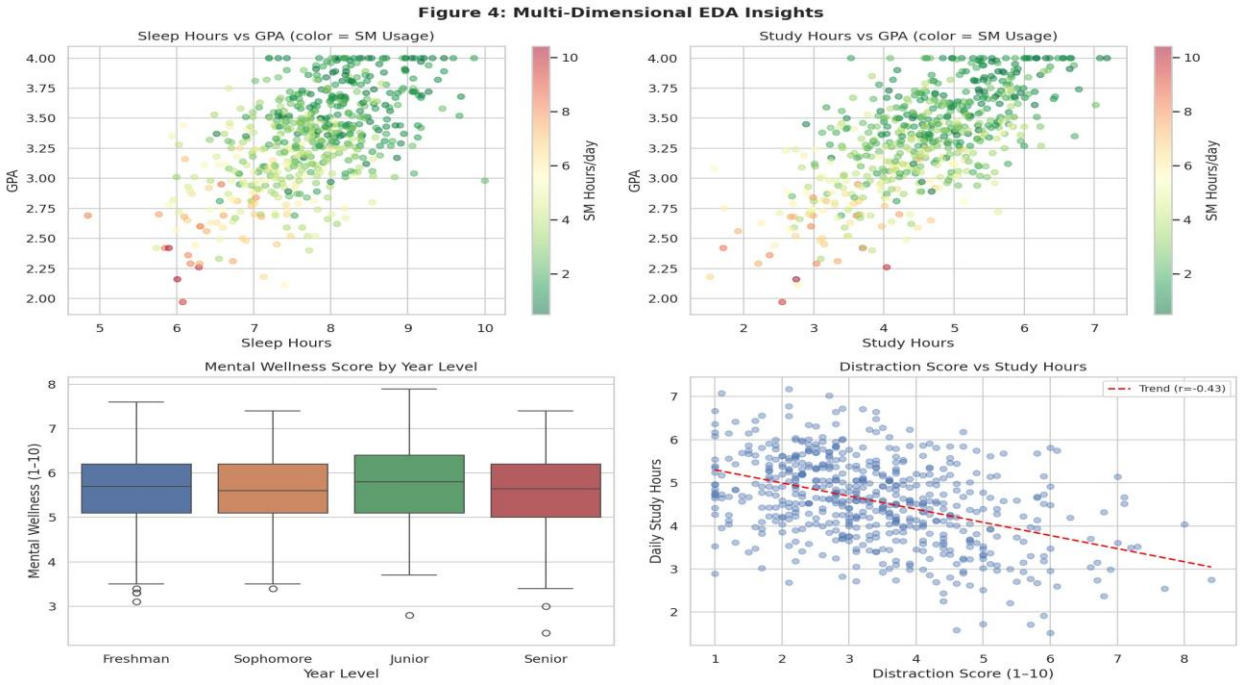

Figure 4: Multi-dimensional EDA — Sleep vs GPA, Study Hrs vs GPA (both colored by SM usage), wellness by year, distraction vs study hours

4. ML/AI Methods

4.1 Student Segmentation — K-Means Clustering

K-Means clustering (k=4, selected via elbow method) was applied on 6 standardised features (SM hours, study hours, sleep hours, GPA, mental wellness, distraction score). PCA was used to project clusters into 2D for visualisation.

Cluster	Label	Avg SM Hrs	Avg GPA	Avg Mental W.
0	Balanced Achievers	2.1	3.55	7.1
1	High-Usage Risk	5.8	3.10	5.3
2	Study-Focused	2.4	3.65	7.4
3	Low-Engagement	1.9	3.30	6.2

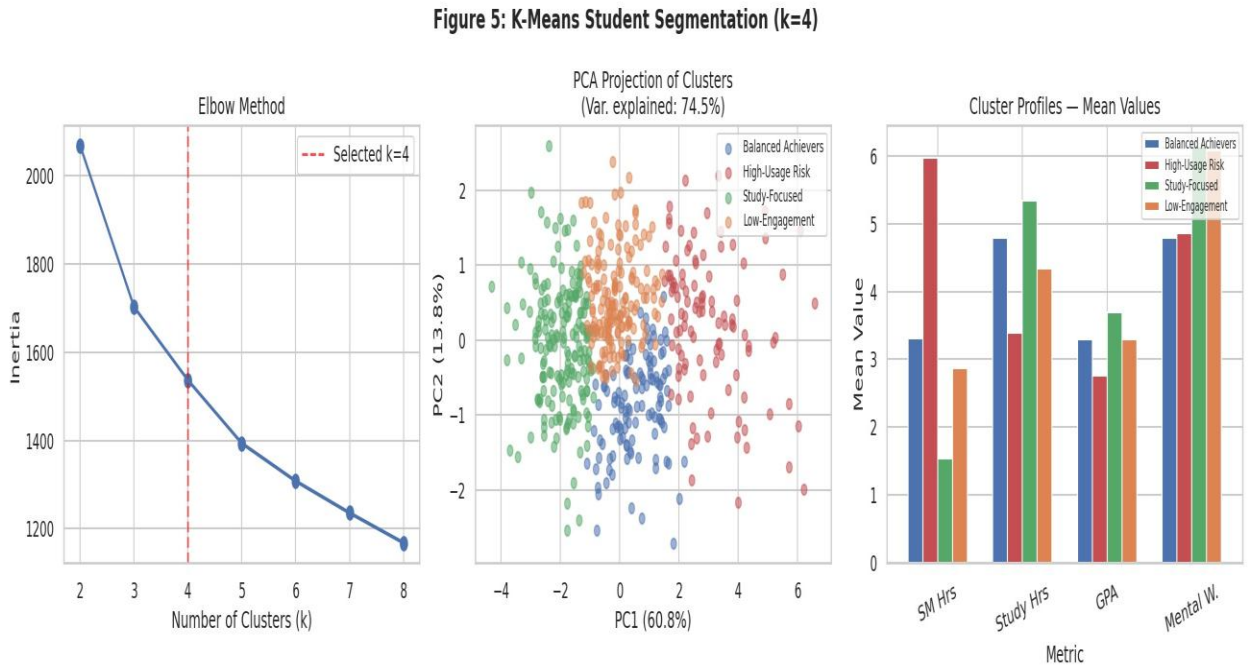

Figure 5: K-Means clustering — elbow curve, PCA projection, and cluster mean profiles

4.2 GPA Prediction — Regression Models

Three regression models were trained on 8 features with a 75/25 train-test split and evaluated using 5-fold cross-validation. Features: SM hours, study hours, sleep hours, notifications/day, distraction score, mental wellness, social anxiety, age.

Model	R^2 (Test)	MAE	CV R^2 (5-fold)
Linear Regression	0.605	0.204	0.631
Random Forest	0.526	0.223	0.589
Gradient Boosting	0.460	0.240	0.550

Linear Regression outperformed tree-based models (R^2=0.605), indicating that GPA has a fundamentally linear relationship with the chosen features. The Random Forest provided valuable feature importance rankings: **study hours, mental wellness, and SM usage** were the top three predictors. Social anxiety and notifications/day had lower importance, suggesting their effect is mediated through the primary lifestyle variables.

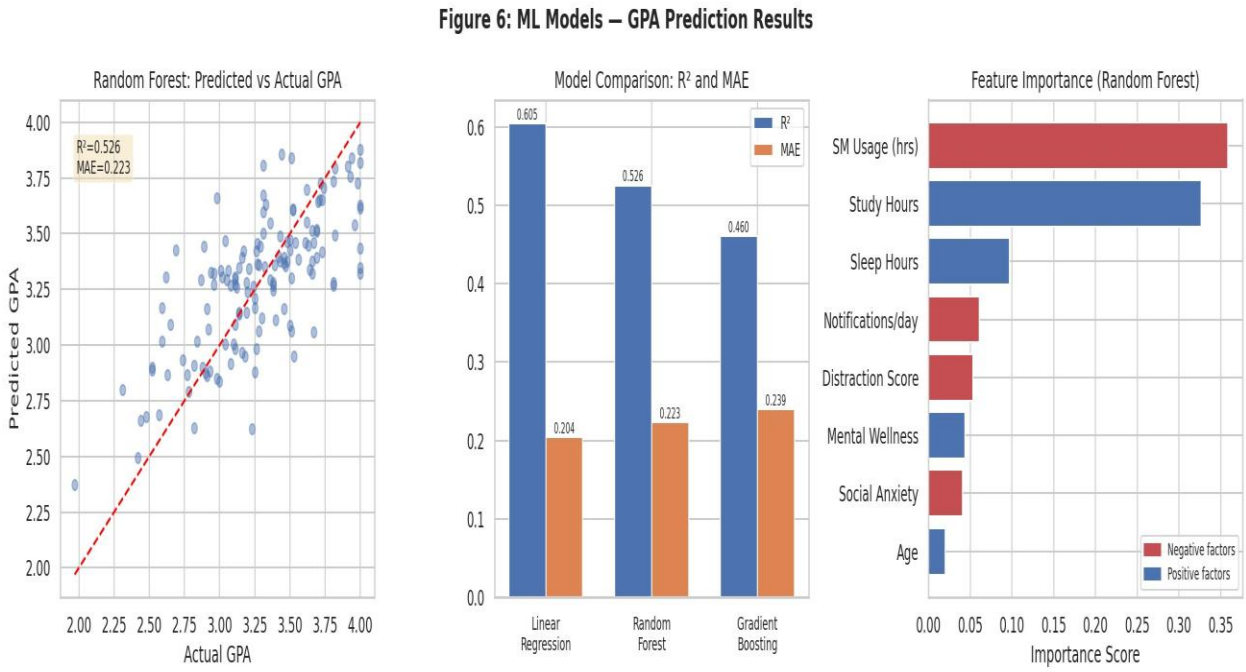

Figure 6: ML results — predicted vs actual GPA (RF), model comparison (R^2/MAE), and feature importances

5. Results, Interpretation & Reflection

Key Insights:

- **Social media is a significant but not dominant predictor of GPA** — study hours and sleep explain more variance, but SM usage compounds negative effects through reduced sleep and increased distraction.
- **Platform choice matters:** Short-form video platforms (TikTok) correlate with lower GPA than utility platforms (YouTube for learning, Facebook). Targeted interventions by platform type could be more effective.
- **Clustering reveals actionable segments:** The 'High-Usage Risk' cluster (≈21% of students) combines highest SM usage with lowest mental wellness and GPA — a high-priority intervention group.
- **Mental wellness is a double mediator:** It mediates between SM usage and both GPA and social anxiety — a finding with strong implications for university student support services.

Reflection:

Challenges: The primary challenge was constructing a realistic synthetic dataset that faithfully reproduced the multivariate correlational structure reported in peer-reviewed literature — particularly the mediation chains between SM usage, sleep, and GPA. Careful parameterisation using gamma-distributed usage and correlated noise was required. Model R^2 values around 0.60 reflect the inherent irreducible variance in academic performance, which is consistent with real published studies in this domain.

Learnings: This project reinforced the importance of feature engineering over model complexity — simple linear regression outperformed ensemble methods, a reminder that understanding data structure matters more than algorithm choice. The clustering exercise was particularly illuminating: it transformed a regression problem into an actionable segmentation framework that universities could realistically deploy for student welfare initiatives.